

Midterm Review Guide

Everything from **Weeks 1 through 6** — cells and genetics, fluids and acid–base, immunity and inflammation, oncology, the two neurological units, and the endocrine system — reduced to the mechanisms, comparisons, and lab patterns the midterm actually tests.

Dr. Vidya Aggrawal

Weeks 1 – 6

6 units + integration

100+ practice questions

Cheat sheets & 30-Q rehearsal exam

HOW THIS EXAM THINKS

The midterm asks the same five shapes the final does. Learn the shapes and unfamiliar questions stop being unfamiliar.

1. **Mechanism:** "Which process explains..." — they want the *step*, not the disease name.
2. **Discriminator:** two look-alikes, one clue separates them (SIADH vs DI → is the urine concentrated or dilute?).
3. **Lab signature:** a pattern of arrows identifies the disorder (↑TSH + ↓T4 = primary hypothyroidism).
4. **Consequence:** given the lesion, predict the downstream effect (no cortisol → no stress response → shock).
5. **Red flag:** which finding means *act now* (Cushing triad, blown pupil, neutropenic fever).

Week 1

Foundations — cell, genetics, fluids, ABGs

Week 2

Immunity — hypersensitivity is the heavy hitter

Week 3

Oncology — benign vs malignant, staging

Week 4

Neuro I — ICP and stroke are guaranteed

Week 5

Neuro II — meningitis, NMJ, cord injury

Week 6

Endocrine — heaviest lab-signature content

WEEK 1

Foundations — Cell, Genetics, Fluids & Acid–Base

The vocabulary the rest of the course is written in: how cells adapt and die, how traits are inherited, and how the body defends its pH.

1 Cell Injury, Death & Adaptation

Adaptation — the cell changes to survive

Adaptation	What changes	Example
Atrophy	Decrease in cell size	Disused muscle after immobilisation; brain in ageing
Hypertrophy	Increase in cell SIZE (no new cells)	Cardiac muscle in hypertension; skeletal muscle in training
Hyperplasia	Increase in cell NUMBER	Endometrium under oestrogen; liver regeneration
Metaplasia	One mature cell type replaced by another	Squamous → columnar in the oesophagus with chronic reflux
Dysplasia	Disordered, atypical growth — not a true adaptation	Cervical dysplasia; the step before carcinoma

Key idea — hypertrophy vs hyperplasia. Hypertrophy = bigger cells. Hyperplasia = more cells. Tissues that cannot divide (cardiac muscle, skeletal muscle, neurons) can **only** hypertrophy — which is why a pressure-overloaded heart thickens rather than growing extra myocytes.

Pearl — metaplasia is adaptive but risky. The substituted cell type is better suited to the stress, but it is abnormal for that location. Sustained long enough, metaplasia → **dysplasia** → carcinoma. That sequence reappears in Week 3.

Reversible vs irreversible injury

Hypoxia → **anaerobic metabolism** → ATP falls and **lactic acid** accumulates → the **Na⁺/K⁺ pump fails** → sodium and water enter → **cellular swelling** (still reversible) → membrane and mitochondrial damage, calcium influx, enzyme release → **irreversible**.

Key idea — ATP is the pivot. Almost every feature of early cell injury traces back to failing ATP: the pump stops, so the cell cannot control its own volume and swells. **Cellular swelling is the earliest visible change** and it is still reversible.

Necrosis vs apoptosis

	NECROSIS	APOPTOSIS
Trigger	Injury — hypoxia, toxins, infection, trauma	Programmed — physiologic or signalled
Cell fate	Swells and ruptures , spilling contents	Shrinks , fragments into tidy apoptotic bodies
Inflammation	Yes — spilled contents provoke it	No — fragments are quietly phagocytosed
Energy	Passive; ATP has already failed	Requires ATP — it is an active process
Scale	Groups of cells	Single scattered cells

MNEMONIC Necrosis is a Mess (swelling, rupture, inflammation). **Apoptosis is Neat** (shrink, package, no inflammation). If the stem mentions surrounding inflammation, it is necrosis.

PRACTICE

Q1. Which change occurs earliest in reversible hypoxic cell injury?

- A. Cellular swelling from failure of the sodium-potassium pump
- B. Nuclear fragmentation
- C. Rupture of lysosomes
- D. Calcification of the cell

A is correct. Falling ATP disables the pump, so sodium and water accumulate inside. Swelling is the earliest visible change and is still reversible; the others indicate irreversible injury.

PRACTICE

Q2. A hypertensive patient's left ventricular wall thickens. Which adaptation is this, and why not the other?

- A. Hypertrophy, because cardiac myocytes cannot divide to increase in number
- B. Hyperplasia, because more myocytes are produced
- C. Metaplasia, because the tissue type changes
- D. Dysplasia, because growth is disordered

A is correct. Mature cardiac muscle is a permanent tissue that cannot undergo mitosis, so the only way to increase workload capacity is for existing cells to enlarge.

PRACTICE

Q3. Which feature distinguishes apoptosis from necrosis?

- A. Apoptosis is energy-dependent and does not provoke inflammation
- B. Apoptosis causes cellular swelling and rupture
- C. Apoptosis affects large groups of adjacent cells
- D. Apoptosis always results from hypoxia

A is correct. Apoptosis is an orderly, ATP-requiring self-destruct programme; the cell packages itself for phagocytosis so nothing spills and no inflammation follows.

2 Genetics & Inheritance

Pattern	How to recognise it	Examples
Autosomal dominant	One copy is enough. Appears in every generation ; an affected parent is usually visible; males and females equally	Huntington disease, Marfan syndrome, familial hypercholesterolaemia
Autosomal recessive	Two copies needed. Carriers are silent , so it can skip generations ; consanguinity raises risk	Cystic fibrosis, sickle cell disease, phenylketonuria
X-linked recessive	Males affected far more often ; passed from carrier mothers; no father-to-son transmission	Haemophilia A, Duchenne muscular dystrophy, red-green colour blindness
Chromosomal	Whole-chromosome gain or loss (nondisjunction)	Trisomy 21 (Down), Turner (45,X), Klinefelter (47,XXY)
Multifactorial	Genes plus environment; clusters in families without a clean pattern	Type 2 diabetes, hypertension, neural tube defects

Key idea — two carrier parents. Autosomal recessive: each pregnancy carries a **25% chance affected, 50% carrier, 25% unaffected non-carrier**. Because carriers show nothing, the first affected child is often the family's first clue — which is exactly why the disease appears to "skip" generations.

Pearl — why X-linked recessive hits males. Males have only one X, so a single defective allele is expressed with nothing to mask it. Females need two copies, so they are usually carriers. And a father gives his son a **Y**, never his X — hence no father-to-son transmission.

Key idea — epigenetics. Gene *expression* can change without altering the DNA sequence — through methylation and histone modification, influenced by diet, stress, and toxins. It explains why identical twins diverge over time.

PRACTICE

Q1. Two phenotypically normal parents have a child with cystic fibrosis. What is the inheritance pattern and the recurrence risk?

- A. Autosomal recessive; 25% with each pregnancy
- B. Autosomal dominant; 50% with each pregnancy
- C. X-linked recessive; 50% of sons
- D. Chromosomal nondisjunction; not recurrent

A is correct. Unaffected parents producing an affected child indicates both are carriers of a recessive allele. Each pregnancy independently carries a one-in-four risk.

PRACTICE

Q2. Which finding argues AGAINST X-linked recessive inheritance?

- A. An affected father has an affected son
- B. Affected males are more common than affected females
- C. The trait passes through unaffected carrier mothers
- D. Maternal uncles are also affected

A is correct. A father passes a Y chromosome to his sons, so an X-linked allele cannot go father to son. The other three are classic X-linked recessive features.

PRACTICE

Q3. Which mechanism produces trisomy 21?

- A. Nondisjunction – failure of chromosomes to separate during meiosis
- B. A single base substitution in a structural gene
- C. Deletion of a tumour suppressor gene
- D. Abnormal methylation of an imprinted region

A is correct. When a chromosome pair fails to separate, one gamete receives an extra copy. Risk rises with maternal age because oocytes have been arrested in meiosis for decades.

3 Fluids, Electrolytes & Oedema

The four causes of oedema

Mechanism	Why fluid leaves the vessel	Example
↑ Capillary hydrostatic pressure	Too much push outward	Heart failure, venous obstruction, fluid overload
↓ Plasma oncotic pressure	Too little pull inward — low albumin	Liver failure, nephrotic syndrome, malnutrition
↑ Capillary permeability	Leaky wall	Inflammation, sepsis, burns, allergy
Lymphatic obstruction	No drainage	Post node dissection, tumour, filariasis

Key idea — albumin is the pull. Albumin generates most plasma oncotic pressure. Anything that lowers it — a liver that cannot synthesise it, a kidney that spills it, or malnutrition — produces oedema by the same mechanism.

The electrolytes worth knowing cold

Electrolyte	Too high	Too low
Sodium (135–145)	Thirst, dry mucous membranes, restlessness, seizures	Confusion, headache, seizures , cerebral oedema
Potassium (3.5–5.0)	Peaked T waves , wide QRS, muscle weakness, cardiac arrest	Flat T waves, U waves, weakness, dysrhythmias, ileus
Calcium (8.5–10.5)	Bones, stones, groans, moans — lethargy, constipation	Tetany , Chvostek and Trousseau signs, laryngospasm

Pearl — potassium and pH move opposite. In **acidosis**, hydrogen ions enter cells and potassium is pushed out → serum potassium **rises**. In **alkalosis** the reverse happens. So a patient in DKA can have a high serum potassium while being *totally* depleted of body potassium.

RED FLAG — CORRECTING SODIUM

Severe hyponatremia with seizures needs treatment, but correcting sodium **too quickly** causes osmotic demyelination. Fast fall in sodium causes cerebral oedema; fast rise causes demyelination — the rate matters as much as the number.

PRACTICE

Q1. A patient with cirrhosis develops generalised oedema with normal cardiac function. Which mechanism is primarily responsible?

- A. Reduced plasma oncotic pressure from impaired albumin synthesis
- B. Increased capillary hydrostatic pressure from fluid overload
- C. Increased capillary permeability from inflammation
- D. Lymphatic obstruction

A is correct. A failing liver cannot make albumin, so the inward pull that normally keeps fluid in the vessel is lost and fluid stays in the interstitium.

PRACTICE

Q2. Which ECG change is the classic early warning of hyperkalemia?

- A. Tall peaked T waves
- B. Prominent U waves
- C. ST elevation
- D. Shortened PR interval

A is correct. Rising extracellular potassium changes the resting membrane potential and speeds repolarisation, producing peaked T waves before the QRS widens toward arrest.

PRACTICE

Q3. Why does acidosis raise the serum potassium?

- A. Hydrogen ions move into cells and potassium shifts out in exchange
- B. The kidneys stop excreting potassium entirely
- C. Potassium intake increases during acidosis
- D. Aldosterone secretion rises sharply

A is correct. It is a transcellular shift, not a change in total body potassium “ which is why treating the acidosis can unmask a profound deficit.

4 Acid-Base & ABGs — a 4-Step Method



More $\text{CO}_2 \rightarrow$ more acid $\rightarrow$ **lower pH**. Blow off $\text{CO}_2 \rightarrow$ less acid $\rightarrow$ **higher pH**.

Value	Normal	Controlled by
pH	7.35 – 7.45	—
PaCO ₂	35 – 45 mmHg	Lungs (fast — minutes)
HCO ₃ [−]	22 – 26 mEq/L	Kidneys (slow — hours to days)
PaO ₂	80 – 100 mmHg	Oxygenation — read separately

THE 4-STEP METHOD

1. **Look at pH.** <7.35 acidosis, >7.45 alkalosis. In range with abnormal values = **fully compensated** (use which side of 7.40).
2. **Look at PaCO₂.** Moves **opposite** to pH $\rightarrow$ **respiratory**.
3. **Look at HCO₃.** Moves the **same** way as pH $\rightarrow$ **metabolic**.
4. **Compensation.** Other value normal = uncompensated. Abnormal but pH still off = partial. pH normal with both abnormal = full.

MNEMONIC ROME — Respiratory Opposite, Metabolic Equal.

Disorder	pH	CO ₂	HCO ₃	Common causes
Respiratory acidosis	↓	↑	n→↑	Hypoventilation: COPD, opioids, neuromuscular weakness, airway obstruction
Respiratory alkalosis	↑	↓	n→↓	Hyperventilation: anxiety, pain, fever, early sepsis, altitude
Metabolic acidosis	↓	n→↓	↓	DKA , lactic acidosis (shock), renal failure, diarrhoea
Metabolic alkalosis	↑	n→↑	↑	Vomiting , NG suction, diuretics, excess antacids

Pearl — vomiting vs diarrhoea. Vomiting loses gastric **acid** → **alkalosis**. Diarrhoea loses intestinal **bicarbonate** → **acidosis**. Same system, opposite disorders, decided by which end the fluid leaves from.

CROSS-UNIT THREAD Full compensation implies a **chronic** process, because the kidneys need hours to days. An ABG that is fully compensated is telling you about time as well as chemistry.

PRACTICE

Q1. ABG: pH 7.28, PaCO₂ 58, HCO₃ 25. What is this?

- A. Uncompensated respiratory acidosis
- B. Compensated respiratory acidosis
- C. Metabolic acidosis
- D. Respiratory alkalosis

A is correct. Low pH with a high CO₂ moving opposite to it = respiratory. Bicarbonate is still normal, so compensation has not begun — an acute picture.

PRACTICE

Q2. A patient with severe diarrhoea for three days is most likely to show which disturbance?

- A. Metabolic acidosis from bicarbonate loss
- B. Metabolic alkalosis from acid loss
- C. Respiratory acidosis
- D. Respiratory alkalosis

A is correct. Intestinal secretions are bicarbonate-rich, so losing them from below depletes buffer and drops the pH. Vomiting does the opposite.

PRACTICE

Q3. ABG: pH 7.37, PaCO₂ 60, HCO₃⁻ 34 in a patient with long-standing lung disease. How is this best described?

- A. Fully compensated respiratory acidosis
- B. Uncompensated metabolic alkalosis
- C. Partially compensated metabolic acidosis
- D. A normal gas

A is correct. The pH is back in range but on the acidic side of 7.40, with both values markedly abnormal – the kidneys have had time to retain bicarbonate, which marks this as chronic.

WEEK 2

Immunity & Inflammation

Two defence systems, the inflammatory response they share, and the four ways immunity turns against the host.

5 Innate vs Adaptive Defence

	INNATE (natural)	ADAPTIVE (acquired)
Speed	Immediate — minutes to hours	Slow first time — days; fast on re-exposure
Specificity	Non-specific — same response to everything	Highly specific to one antigen
Memory	None — identical every time	Yes — the basis of vaccination
Players	Skin and mucosa, normal flora, neutrophils , macrophages, NK cells, complement, fever, inflammation	B cells → antibodies (humoral); T cells → cell-mediated

The adaptive arms

- **Humoral (B cells → plasma cells → antibodies)** — handles **extracellular** organisms and toxins.
- **Cell-mediated (T cells)** — handles **intracellular** threats: viruses, some bacteria, tumour cells, and transplanted tissue. **CD4 helper** T cells coordinate; **CD8 cytotoxic** T cells kill.

Antibody	Remember it as
IgM	M = iMMediate — first antibody in a new infection; largest
IgG	G = Greatest — most abundant, memory response, only one crossing the placenta
IgA	A = Areas of secretion — mucosa, saliva, tears, breast milk
IgE	E = Eosinophils/allergy — type I hypersensitivity and parasites
IgD	D = Doorway — B-cell surface receptor

Key idea — active vs passive immunity. **Active** = your own immune system responds and builds memory (infection or vaccination) — slow but lasting. **Passive** = you receive ready-made antibodies (placenta, breast milk, immunoglobulin injection) — immediate but temporary, with **no memory**.

Pearl — why the newborn is protected then vulnerable. Maternal **IgG** crosses the placenta and **IgA** arrives in breast milk — both passive. That protection fades over the first months, which is precisely when the vaccination schedule begins.

PRACTICE

Q1. Which feature belongs to adaptive rather than innate immunity?

- A. Immunologic memory allowing a faster response on re-exposure
- B. Physical barriers such as intact skin
- C. Neutrophil phagocytosis
- D. The fever response

A is correct. Memory is the defining feature of adaptive immunity and the reason vaccines work. The others are innate and behave identically at every exposure.

PRACTICE

Q2. Which immunoglobulin crosses the placenta to protect the newborn?

- A. IgG
- B. IgM
- C. IgA
- D. IgE

A is correct. IgG is the only isotype that crosses the placenta, giving passive protection for the first months. IgA arrives separately through breast milk.

PRACTICE

Q3. A patient receives immunoglobulin after a needlestick. What kind of immunity is this?

- A. Passive – immediate protection with no memory
- B. Active – the patient produces their own antibodies
- C. Innate – a non-specific barrier defence
- D. Cell-mediated – T cells are transferred

A is correct. Ready-made antibodies act at once but are consumed and not replaced, so there is no lasting protection – the trade-off of passive immunity.

6 Inflammation & Healing

The five cardinal signs and why each happens

Sign	Mechanism
Redness (rubor)	Vasodilation brings more blood to the area
Heat (calor)	Same increased blood flow
Swelling (tumor)	Increased capillary permeability → fluid and protein leak into tissue
Pain (dolor)	Pressure from swelling plus mediators (bradykinin, prostaglandins)
Loss of function	Pain and swelling together

Key idea — the whole picture comes from two vascular changes. Vasodilation (redness, heat) and **increased permeability** (swelling, and therefore pain). Histamine from **mast cells** is the first mediator; neutrophils are the first cells to arrive.

Acute vs chronic

	ACUTE	CHRONIC
Onset/duration	Immediate, days	Weeks to years
Dominant cell	Neutrophils	Macrophages and lymphocytes
Outcome	Resolution, or progression to chronic	Fibrosis, scarring, granuloma formation

Pearl — granulomas. When macrophages cannot destroy something (TB, foreign material), they wall it off into a granuloma. A granuloma is the immune system saying "I cannot kill this, so I will contain it."

Healing

1. **Haemostasis** — vasoconstriction and clot.
2. **Inflammation** — neutrophils then macrophages clear debris.

3. **Proliferation** — fibroblasts lay collagen, angiogenesis, granulation tissue, epithelialisation.
4. **Remodelling** — collagen reorganises; scar reaches only about **80%** of original strength.

Key idea — what delays healing: poor perfusion, infection, **diabetes** (impaired leukocytes plus microvascular disease), **corticosteroids** (suppress inflammation and collagen synthesis), smoking, malnutrition (**protein, vitamin C, zinc**), advanced age, and unrelieved pressure.

PRACTICE

Q1. Which vascular change directly produces the swelling of acute inflammation?

- A. Increased capillary permeability allowing fluid and protein into tissue
- B. Vasoconstriction of arterioles
- C. Increased lymphatic drainage
- D. Decreased capillary hydrostatic pressure

A is correct. Histamine widens the junctions between endothelial cells so fluid and protein escape. Vasodilation explains redness and heat; permeability explains swelling.

PRACTICE

Q2. Which cell predominates in chronic inflammation?

- A. Macrophages and lymphocytes
- B. Neutrophils
- C. Eosinophils
- D. Platelets

A is correct. Neutrophils dominate the acute phase; when the stimulus persists, macrophages and lymphocytes take over and drive fibrosis and granuloma formation.

PRACTICE

Q3. Why do corticosteroids impair wound healing?

- A. They suppress the inflammatory phase and inhibit fibroblast collagen synthesis
- B. They increase capillary permeability excessively
- C. They accelerate epithelialisation too quickly
- D. They raise the tissue oxygen requirement

A is correct. Inflammation is a necessary stage of healing, not merely a nuisance. Suppressing it and the collagen deposition that follows leaves a weaker, slower-healing wound.

7 The Four Hypersensitivities

Type	Mediator	Timing	Mechanism	Examples
I — Immediate	IgE + mast cells	Minutes	Allergen cross-links IgE → histamine release → vasodilation, permeability, bronchoconstriction	Anaphylaxis , hay fever, allergic asthma, food and bee-sting allergy
II — Cytotoxic	IgG/IgM against a specific cell	Hours	Antibody binds host tissue → complement and phagocytes destroy that cell	ABO transfusion reaction , haemolytic disease of the newborn, Graves, myasthenia gravis
III — Immune complex	Antigen–antibody complexes	Hours to days	Complexes deposit in vessels, joints, glomeruli → complement-driven inflammation	SLE , rheumatoid arthritis, post-streptococcal glomerulonephritis, serum sickness
IV — Delayed	T cells — no antibody	24–72 hours	Sensitised T cells and macrophages attack directly	Poison ivy / contact dermatitis , TB skin test, transplant rejection, latex

MNEMONIC ACID — **A**llergic (I), **C**ytotoxic (II), **I**mmune complex (III), **D**elayed (IV). And: type **IV** is the only one that is cell-mediated with **no antibody**.

Key idea — sensitisation must happen first. The **first** exposure produces IgE and is uneventful. The **second** exposure cross-links that IgE on mast cells and triggers the reaction. This is why a patient can eat a food safely for years and then react violently.

RED FLAG — ANAPHYLAXIS

Widespread mast-cell degranulation causes **airway oedema and bronchoconstriction** plus massive **vasodilation and capillary leak** → **distributive shock**. Facial swelling, stridor, wheeze, and hypotension within minutes of an exposure demand **epinephrine immediately** — it reverses both the airway and the vascular problem.

PRACTICE

Q1. A gardener develops an itchy vesicular rash 48 hours after contact with poison ivy. Which hypersensitivity is this?

- A. Type IV "delayed, T-cell mediated"
- B. Type I "IgE and mast cells"
- C. Type II "antibody against tissue"
- D. Type III "immune complex deposition"

A is correct. The 24-to-72-hour delay is the signature of a T-cell mediated reaction. Type I would occur within minutes.

PRACTICE

Q2. An ABO-incompatible transfusion causes red cell destruction. Which mechanism is responsible?

- A. Type II "antibodies bind donor red cells and trigger complement-mediated lysis"
- B. Type I "IgE-mediated mast cell degranulation"
- C. Type III "circulating immune complexes deposit in tissue"
- D. Type IV "cytotoxic T cells attack the donor cells"

A is correct. Recipient antibodies target antigens on a specific cell type, which is the defining feature of type II. The result is intravascular haemolysis with fever, flank pain, and dark urine.

PRACTICE

Q3. Why is the first exposure to an allergen usually uneventful?

- A. It only sensitises – IgE is produced and binds mast cells, with no mediator release yet
- B. The allergen is destroyed before reaching the immune system
- C. Mast cells are absent until the second exposure
- D. Complement is not activated on first contact

A is correct. Sensitisation arms the mast cells but releases nothing. Only re-exposure cross-links the bound IgE and triggers degranulation – which is why reactions often appear "out of nowhere."

8 Autoimmunity & Immunodeficiency

Key idea — two opposite failures. Autoimmunity = loss of **self-tolerance**; the immune system attacks the host. **Immunodeficiency** = the immune system fails to attack anything; the problem is **infection**, especially by organisms that are normally harmless.

Condition	Target	Signature
Systemic lupus erythematosus	Nuclear antigens; immune complexes (type III)	Butterfly rash , photosensitivity, joint pain, nephritis; ANA positive; remits and flares
Rheumatoid arthritis	Synovium	Symmetric small-joint pain, morning stiffness >1 hour , systemic symptoms
Type 1 diabetes	Pancreatic beta cells	Absolute insulin deficiency; presents in youth, ketosis-prone
Graves / Hashimoto	Thyroid (stimulating vs destroying)	Hyper- vs hypothyroidism — see Week 6
Myasthenia gravis	Acetylcholine receptors	Fatigable weakness — see Week 5

Immunodeficiency	Mechanism	Consequence
Primary (congenital)	Inherited defect in a component of immunity	Recurrent infection from infancy
HIV / AIDS	Virus destroys CD4 helper T cells	Adaptive coordination collapses → opportunistic infections and unusual malignancies
Iatrogenic	Chemotherapy, corticosteroids, transplant immunosuppression	Neutropenia — fever may be the only sign of serious infection

RED FLAG — NEUTROPENIC FEVER

Without neutrophils there is **no pus and minimal inflammation**, so the usual localising signs never appear. In a chemotherapy patient, **fever alone** may be the only evidence of overwhelming infection and is treated as an emergency.

Pearl — why CD4 count matters in HIV. The CD4 helper T cell is the conductor of the adaptive response. Destroy it and both the antibody and cytotoxic arms lose coordination — which is why the infections that appear are ones a healthy immune system dispatches without notice.

PRACTICE

Q1. Which statement best describes autoimmunity?

- A. Loss of self-tolerance, so the immune system attacks the body's own tissue
- B. Failure of the immune system to respond to any antigen
- C. An exaggerated response to a harmless environmental antigen
- D. Destruction of CD4 lymphocytes by a virus

A is correct. Option C describes allergy and option D describes HIV. Autoimmunity is specifically the breakdown of the ability to distinguish self from non-self.

PRACTICE

Q2. Why can a serious infection present with fever alone in a neutropenic patient?

- A. Without neutrophils there is no pus or local inflammation to produce localising signs
- B. Fever suppresses all other symptoms
- C. Neutropenia prevents bacteria from multiplying
- D. Antibiotics mask the usual findings

A is correct. The visible signs of infection are largely produced by neutrophils arriving at the site. With none available, infection can be advanced while looking unimpressive.

PRACTICE

Q3. Systemic lupus erythematosus damages the kidney and joints primarily through which mechanism?

- A. Type III “immune complexes deposit in tissue and trigger complement-mediated inflammation
- B. Type I “IgE-mediated mast cell degranulation
- C. Type IV “delayed T-cell mediated attack
- D. Direct viral cytotoxicity

A is correct. Lupus generates antibodies against nuclear antigens; the resulting complexes lodge in small vessels, glomeruli, and joints, causing widespread inflammation.

WEEK 3

Oncology — The Biology of Cancer

What makes a tumour malignant, the genes behind it, how it spreads, and why cancer makes the whole body sick.

9 Benign vs Malignant

	BENIGN	MALIGNANT
Growth rate	Slow	Rapid, unpredictable
Differentiation	Well differentiated — resembles the parent tissue	Poorly differentiated / anaplastic — may not resemble it at all
Border	Encapsulated , well defined, mobile	Invasive , irregular, fixed to surrounding tissue
Metastasis	Never	Yes — the defining feature
Recurrence	Rare after excision	Common
Systemic effect	Local only — harmful by pressure or location	Cachexia , weight loss, fever, night sweats

Key idea — two features define malignancy: *invasion* and *metastasis*. Everything else on the table is a correlate. A tumour that stays put and stays encapsulated is benign no matter how fast it grows.

Pearl — benign does not mean harmless. A benign meningioma in a fixed skull, or a benign tumour compressing an airway, can kill by **location** alone. "Benign" describes behaviour, not consequence.

Naming

Suffix	Meaning	Example
-oma	Usually benign	Lipoma (fat), leiomyoma (smooth muscle), adenoma (gland)
-carcinoma	Malignant, from epithelium	Adenocarcinoma, squamous cell carcinoma. Most common cancers
-sarcoma	Malignant, from connective tissue	Osteosarcoma, liposarcoma
-blastoma	Malignant, from embryonic/precursor cells	Neuroblastoma, retinoblastoma — typically paediatric

MNEMONIC Carcinoma = **Covering** (epithelium). Sarcoma = **Support** (connective tissue).

Exceptions to the "-oma is benign" rule that are worth knowing: **melanoma, lymphoma, myeloma** are all malignant.

PRACTICE

Q1. Which single feature most reliably defines a tumour as malignant?

- A. The ability to invade surrounding tissue and metastasise
- B. A rapid rate of growth
- C. A large size at diagnosis
- D. Causing pain or obstruction

A is correct. Benign tumours can be large, fast-growing, and dangerous by location, but they never invade or metastasise. That capability is what defines malignancy.

PRACTICE

Q2. A breast lump is hard, irregular, fixed, and painless, with overlying skin dimpling. What do these features suggest?

- A. Malignancy – fixation and irregularity indicate local invasion
- B. A benign fibroadenoma
- C. A simple cyst
- D. Acute mastitis

A is correct. Benign masses tend to be smooth, mobile, well defined, and often tender. Hard, fixed, irregular, and painless with skin change is the malignant pattern.

PRACTICE

Q3. Osteosarcoma arises from which tissue type?

- A. Connective tissue – the suffix -sarcoma indicates mesenchymal origin
- B. Epithelium lining a gland
- C. Lymphoid tissue
- D. Embryonic precursor cells

A is correct. Sarcomas arise from connective and supporting tissues – bone, cartilage, muscle, fat. Carcinomas arise from epithelium and are far more common overall.

10 Carcinogenesis & the Genes

Initiation (a carcinogen causes an irreversible DNA mutation) → **Promotion** (repeated exposure drives the mutated clone to proliferate — this stage is *reversible*) → **Progression** (further mutations bring invasion and metastasis).

Gene type	Normal job	What goes wrong	Analogy
Proto-oncogene → oncogene	Promotes normal, controlled growth	Gain of function — one mutated copy is enough	A stuck accelerator
Tumour suppressor gene	Restraints division; repairs DNA; triggers apoptosis	Loss of function — usually both copies must be lost	A failed brake
DNA repair genes	Correct replication errors	Loss allows mutations to accumulate everywhere	A broken spellchecker

Key idea — p53 is the most important tumour suppressor. It halts the cell cycle for repair and triggers apoptosis if damage is irreparable, which is why it is called the guardian of the genome. Lose it and damaged cells survive and divide. **BRCA1/2** are DNA-repair genes; inheriting one faulty copy means only one more hit is needed.

Pearl — why cancer takes years. A single mutation is not enough. Multiple hits must accumulate in the same cell lineage — which explains the long latency after carcinogen exposure and why incidence rises steeply with age.

Carcinogens and risk

- **Chemical** — tobacco smoke (the single largest preventable cause), asbestos, benzene, alcohol.
- **Physical** — ionising radiation, **UV light** (skin cancers).
- **Biological** — **HPV** (cervical), hepatitis B and C (hepatocellular), **EBV**, *H. pylori* (gastric).
- **Host factors** — age, inherited mutations, immunosuppression, chronic inflammation.

CROSS-UNIT THREAD Chronic inflammation is carcinogenic. Sustained tissue injury and repair means sustained cell division, and every division is a chance for an uncorrected mutation. This connects Week 2 to Week 3 — and explains reflux → Barrett → adenocarcinoma, and *H. pylori* → gastritis → gastric cancer.

PRACTICE

Q1. How does the loss of a tumour suppressor gene differ from the activation of an oncogene?

- A. Tumour suppressors usually require both copies to be lost, while one activated oncogene copy is sufficient
- B. Tumour suppressor loss requires only one copy, while oncogenes require both
- C. Both require the loss of both copies
- D. Neither is inherited

A is correct. A brake still works if one of two is intact, so both must fail. An accelerator only needs to jam once " which is why oncogene mutations act dominantly.

PRACTICE

Q2. Which function of p53 explains why its loss is so consequential?

- A. It arrests the cell cycle for DNA repair and triggers apoptosis when damage cannot be fixed
- B. It directly stimulates cell division
- C. It forms new blood vessels for growing tissue
- D. It transports oxygen to dividing cells

A is correct. p53 is the checkpoint that either fixes damage or eliminates the cell. Without it, genetically damaged cells survive, divide, and accumulate further mutations.

PRACTICE

Q3. Which stage of carcinogenesis is potentially reversible?

- A. Promotion – removing the promoting agent can halt clonal expansion
- B. Initiation – the DNA mutation can repair itself spontaneously
- C. Progression – invasion can regress without treatment
- D. Metastasis

A is correct. Initiation is an irreversible mutation, but promotion depends on continued exposure – which is the biological reason smoking cessation lowers risk even after years of smoking.

11 Spread, Staging & Grading

How a tumour metastasises

1. Cells **lose adhesion** to one another.
2. They **invade through the basement membrane** — the step that defines malignancy.
3. **Angiogenesis** — the tumour signals for new vessels to feed itself and to provide an exit.
4. Cells **intravasate** into lymph or blood, travel, and **extravasate** at a distant site.
5. A **secondary tumour** establishes.

Route	Typical of	Where it lands
Lymphatic	Carcinomas	Regional nodes first — why nodes are sampled and why the sentinel node is biopsied
Haematogenous	Sarcomas	Lung, liver, bone, brain — the organs with big capillary beds
Seeding	Ovarian, gastric, colon	Directly across a body cavity (peritoneum)

Key idea — staging vs grading. Staging = how far it has **SPREAD** (TNM: Tumour size, Nodes, Metastasis). Grading = how abnormal the cells **LOOK** down the microscope (differentiation). **Staging drives prognosis and treatment**; grading describes aggressiveness.

MNEMONIC Stage = Spread. Grade = Glance (down a microscope). If the question mentions nodes or distant organs, it is staging.

Pearl — why lung, liver, bone, and brain. They receive enormous blood flow through extensive capillary beds, so circulating tumour cells are filtered out there. Those four are the default answer for "most common metastatic sites."

PRACTICE

Q1. A staging report describes tumour size, nodal involvement, and distant spread. What is being assessed?

- A. Staging – the anatomic extent of disease, using the TNM system
- B. Grading – how differentiated the cells appear
- C. Initiation – the first mutation event
- D. Promotion – clonal expansion

A is correct. TNM is the staging system and it measures spread. Grading is a microscopic assessment of how closely the cells still resemble their tissue of origin.

PRACTICE

Q2. Which step distinguishes a malignant tumour from carcinoma in situ?

- A. Invasion through the basement membrane
- B. Loss of a tumour suppressor gene
- C. An increased rate of mitosis
- D. The presence of angiogenesis

A is correct. Carcinoma in situ has all the cellular features of malignancy but has not breached the basement membrane. Crossing it is what makes spread possible.

PRACTICE

Q3. Why is angiogenesis essential to tumour growth?

- A. Beyond a small size a tumour cannot get enough oxygen and nutrients by diffusion alone
- B. New vessels suppress the immune response
- C. Angiogenesis directly causes mutations
- D. It converts benign tumours into sarcomas

A is correct. Diffusion supports only a millimetre or two of tissue. Recruiting a blood supply lets the tumour keep growing and simultaneously provides the route for metastasis.

12 Systemic & Treatment Effects

Effect	Mechanism
Cachexia — wasting despite intake	Tumour-driven cytokines raise metabolic rate and break down muscle and fat. It is not simply poor appetite
Anaemia	Chronic disease, blood loss, marrow infiltration, or treatment
Fever, night sweats, weight loss	Cytokine release — the "B symptoms" of lymphoma
Paraneoplastic syndromes	Tumour secretes a hormone-like substance: SIADH (small cell lung), PTH-related peptide → hypercalcaemia, ACTH → Cushing
Pain	Pressure, invasion of nerves, obstruction, bone metastasis

Key idea — why chemotherapy causes the side effects it does. It targets **rapidly dividing** cells and cannot tell tumour from healthy tissue. So the tissues that divide fastest suffer: **bone marrow** (anaemia, bleeding, **neutropenia**), **GI mucosa** (mucositis, nausea, diarrhoea), and **hair follicles** (alopecia). Every classic side effect is predictable from that one principle.

RED FLAGS IN ONCOLOGY

Neutropenic fever — fever may be the only sign of overwhelming infection. **Spinal cord compression** — new back pain with leg weakness or urinary retention risks permanent paralysis.

Tumour lysis syndrome — after starting treatment, $\uparrow K^+$, $\uparrow$ phosphate, $\uparrow$ uric acid with $\downarrow$ **calcium** and acute kidney injury. **Hypercalcaemia of malignancy** — confusion, constipation, dehydration.

Pearl — tumour lysis calcium. Everything intracellular goes **up** (potassium, phosphate, uric acid) and calcium goes **down**, because the released phosphate binds it. That inverted calcium is the giveaway.

PRACTICE

Q1. Why does chemotherapy commonly cause neutropenia, mucositis, and alopecia together?

- A. It targets rapidly dividing cells and cannot distinguish tumour from normal fast-turnover tissue
- B. It is directly toxic to the immune system only
- C. It causes an allergic reaction in skin and mucosa
- D. It depletes iron stores

A is correct. Marrow, GI mucosa, and hair follicles are the fastest-dividing normal tissues, so they take collateral damage. One mechanism explains all three side effects.

PRACTICE

Q2. A patient with small cell lung cancer develops hyponatremia with concentrated urine. What is happening?

- A. A paraneoplastic syndrome – the tumour is secreting ADH (SIADH)
- B. Dehydration from poor intake
- C. Diabetes insipidus from a brain metastasis
- D. Tumour lysis syndrome

A is correct. Some tumours secrete hormone-like substances. Ectopic ADH causes water retention, diluting the serum sodium while the urine stays inappropriately concentrated.

PRACTICE

Q3. A cancer patient reports new back pain with leg weakness and difficulty voiding. What is the priority concern?

- A. Spinal cord compression from metastasis – delay risks permanent paralysis
- B. Simple mechanical back strain
- C. Peripheral neuropathy from chemotherapy
- D. Hypercalcaemia of malignancy

A is correct. The combination of back pain with neurologic deficit and bladder involvement is a classic oncologic emergency requiring immediate imaging and treatment.

WEEK 4

Neurologic Structure, Function & Disorders

How the brain is supplied and protected, what happens when pressure rises, and the two great vascular and traumatic injuries.

13 Neurons, Glia & Brain Anatomy

Cell	Role
Neuron	Signal transmission. Cannot regenerate in the CNS — which is why CNS damage is usually permanent
Astrocyte	Supports neurons; forms the blood–brain barrier ; scars after injury
Oligodendrocyte	Makes myelin in the CNS (the target in multiple sclerosis)
Schwann cell	Makes myelin in the PNS (the target in Guillain-Barré)
Microglia	The CNS macrophage — immune defence

Key idea — myelin speeds conduction. Saltatory conduction lets impulses jump between nodes of Ranvier. Demyelination therefore slows or blocks transmission without killing the neuron — producing the fluctuating, sometimes recoverable deficits of MS and Guillain-Barré.

Region	Function	Deficit when damaged
Frontal lobe	Executive function, personality, motor cortex, Broca area	Disinhibition, poor judgement, contralateral weakness, expressive aphasia
Parietal	Sensation, spatial awareness	Neglect, sensory loss
Temporal	Hearing, memory, Wernicke area	Memory loss, receptive aphasia
Occipital	Vision	Visual field loss, cortical blindness
Cerebellum	Coordination, balance	Ataxia , intention tremor, dysmetria
Brainstem	Cardiac, respiratory, arousal centres	Life-threatening — loss of breathing and consciousness

Pearl — Broca vs Wernicke. **Broca = Broken speech** (expressive; the patient knows what they want to say and cannot produce it, and is frustrated). **Wernicke = Word salad** (receptive; fluent but meaningless, and the patient is often unaware).

PRACTICE

Q1. A patient speaks fluently but with meaningless content and seems unaware of the problem. Which area is affected?

- A. Wernicke area in the temporal lobe “ receptive aphasia
- B. Broca area in the frontal lobe “ expressive aphasia
- C. The cerebellum
- D. The occipital lobe

A is correct. Wernicke aphasia is fluent but empty, and comprehension “ including of one's own speech “ is impaired, which is why insight is lost.

PRACTICE

Q2. Which cell produces myelin in the central nervous system?

- A. Oligodendrocyte
- B. Schwann cell
- C. Astrocyte
- D. Microglia

A is correct. Oligodendrocytes myelinate CNS axons and are the target in multiple sclerosis. Schwann cells do the same job in the peripheral nervous system.

PRACTICE

Q3. Why is brainstem injury immediately life-threatening?

- A. It houses the cardiac, respiratory, and arousal centres
- B. It contains the primary motor cortex
- C. It is responsible for coordination and balance
- D. It processes all visual input

A is correct. The brainstem controls the autonomic functions that sustain life, so even small lesions there can stop breathing or abolish consciousness.

14 Cerebral Perfusion & Intracranial Pressure

Key idea — the Monro-Kellie doctrine. The skull is a **rigid box** holding three things: **brain tissue, blood, and CSF**. Their total volume is fixed. If one increases, another must decrease or pressure rises. Early on, CSF and venous blood are displaced and the patient looks fine — then compensation is exhausted and pressure climbs steeply.

Cerebral Perfusion Pressure = Mean Arterial Pressure – Intracranial Pressure

So a rising ICP **directly reduces** brain perfusion. The body's response is to raise systemic blood pressure to force blood in — which is exactly what produces the Cushing triad.

Stage	Findings
Earliest and most sensitive	A change in level of consciousness — restlessness, confusion, drowsiness
Early	Headache, nausea and projectile vomiting (without nausea), pupillary sluggishness, visual changes
Late	Cushing triad: rising systolic BP with widening pulse pressure, BRADYCARDIA, irregular respirations
Herniation	Fixed dilated pupil (usually unilateral first), posturing, coma, respiratory arrest

RED FLAG — CUSHING TRIAD IS LATE

By the time blood pressure rises and the pulse falls, herniation is close. The finding that should prompt action long before this is **a deteriorating level of consciousness**. Also: a **new unilateral fixed dilated pupil** means the third cranial nerve is being compressed by shifting brain — a neurosurgical emergency.

Pearl — why hypercapnia is dangerous here. CO₂ is a potent **cerebral vasodilator**. Rising CO₂ dilates cerebral vessels, adds blood volume to a full box, and raises ICP further. This is why ventilation and oxygenation are managed carefully in head injury — a Week 1 acid–base concept driving a Week 4 emergency.

PRACTICE

Q1. Which is the EARLIEST and most sensitive indicator of rising intracranial pressure?

- A. A change in level of consciousness
- B. The Cushing triad
- C. A fixed dilated pupil
- D. Decerebrate posturing

A is correct. Altered consciousness appears well before the classic late signs. Options B, C, and D all indicate that compensation has already failed.

PRACTICE

Q2. Which set of vital sign changes constitutes the Cushing triad?

- A. Rising blood pressure, falling pulse, irregular respirations
- B. Falling blood pressure, rising pulse, rapid respirations
- C. Rising blood pressure, rising pulse, rapid respirations
- D. Falling blood pressure, falling pulse, shallow respirations

A is correct. The body raises systemic pressure to overcome the intracranial pressure, and reflex bradycardia follows. Option B describes hypovolaemic shock – the opposite pattern.

PRACTICE

Q3. Why does rising carbon dioxide worsen intracranial pressure?

- A. CO₂, dilates cerebral vessels, increasing intracranial blood volume in a fixed space
- B. CO₂, increases CSF production directly
- C. CO₂, causes cerebral vasoconstriction and ischemia
- D. CO₂, damages the blood-brain barrier

A is correct. In a rigid skull, any added volume raises pressure. Because CO₂ is a powerful cerebral vasodilator, letting it climb adds blood volume exactly where there is no room.

15 Stroke & Traumatic Brain Injury

	ISCHEMIC (~85%)	HEMORRHAGIC
Cause	Thrombotic (local atherosclerosis) or embolic (often from atrial fibrillation)	Rupture — hypertension, aneurysm, arteriovenous malformation
Onset	Sudden focal deficit; thrombotic may stutter or occur on waking	Sudden, often with severe headache ("worst of my life"), vomiting, rapid decline
Treatment logic	Restore flow — time-critical	Control bleeding and pressure; thrombolytics are contraindicated

Key idea — the ischemic penumbra. Around the dead core sits tissue that is **stunned but still alive**. Restore perfusion quickly and it recovers; leave it and it is recruited into the infarct. That salvageable rim is the entire reason for the treatment window and for "time is brain."

MNEMONIC BE FAST — **B**alance loss, **E**ye/vision change, **F**ace droop, **A**rm weakness, **S**peech difficulty, **T**ime to call for help.

Key idea — deficits are CONTRALATERAL. Motor and sensory tracts cross, so a **left**-hemisphere stroke produces **right**-sided weakness — and, in most people, aphasia, because language is usually left-hemisphere. A right-hemisphere stroke more often produces neglect and impulsivity.

Pearl — a TIA is a warning, not a small stroke. Deficits resolve fully within 24 hours with no infarction, but the short-term risk of a completed stroke is high — so a TIA is worked up urgently rather than reassured away.

Traumatic brain injury

Type	Feature
Concussion	Transient functional disturbance; brief confusion or loss of consciousness; imaging normal
Contusion	Actual bruising of brain tissue — coup at impact, contrecoup opposite
Epidural haematoma	Arterial (middle meningeal). Classic lucid interval then rapid deterioration — a surgical emergency
Subdural haematoma	Venous, slower. Common in older adults and alcohol use after minor trauma; may present days to weeks later
Diffuse axonal injury	Widespread shearing of axons from acceleration–deceleration; prolonged coma with poor outcome

RED FLAG — THE LUCID INTERVAL

A patient who is knocked out, wakes up and seems fine, then deteriorates rapidly is the classic **epidural haematoma**. The apparent recovery is deceptive — arterial bleeding is filling the space.

PRACTICE

Q1. A right-handed patient has left-sided facial droop and left arm weakness. Where is the lesion?

- A. The right cerebral hemisphere, because motor tracts cross
- B. The left cerebral hemisphere
- C. The cerebellum
- D. The spinal cord at C7

A is correct. Corticospinal tracts decussate, so deficits appear on the side opposite the lesion. Left-sided weakness therefore means a right-hemisphere event.

PRACTICE

Q2. Why is rapid reperfusion the goal in ischemic stroke?

- A. The penumbra surrounding the infarct core is still viable and can be saved
- B. Reperfusion reverses cell death in the core
- C. It prevents the stroke from becoming hemorrhagic
- D. It lowers intracranial pressure directly

A is correct. The core is already lost. The treatment window exists entirely to rescue the surrounding stunned tissue before it dies too.

PRACTICE

Q3. A patient loses consciousness after a head injury, wakes and converses normally, then rapidly deteriorates. What is most likely?

- A. Epidural haematoma with a classic lucid interval
- B. Concussion
- C. Chronic subdural haematoma
- D. Diffuse axonal injury

A is correct. Arterial bleeding accumulates quickly. The lucid interval occurs before the expanding clot exhausts intracranial compensation, after which decline is abrupt.

Disorder	Mechanism	Signature
Parkinson disease	Loss of dopamine -producing neurons in the substantia nigra	Resting tremor (pill-rolling), rigidity (cogwheel), bradykinesia , postural instability, shuffling gait, masked face
Multiple sclerosis	Autoimmune demyelination of CNS white matter	Relapsing-remitting visual, sensory, and motor deficits separated in time and space ; fatigue; heat worsens symptoms
Huntington disease	Autosomal dominant ; loss of inhibitory neurons	Chorea , cognitive decline, psychiatric change; onset in mid-adulthood
Alzheimer disease	Amyloid plaques and neurofibrillary tangles	Progressive memory loss beginning with recent memory

Pearl — tremor timing. A **resting** tremor that improves with movement points to **Parkinson**. An **action/intention** tremor that worsens as the hand approaches a target points to the **cerebellum** or essential tremor.

Seizures

- **Focal** — begins in one area; may or may not impair awareness.
- **Generalised** — both hemispheres from the start; includes **tonic-clonic** and absence.
- **Status epilepticus** — prolonged or repeated seizures without recovery between. A **medical emergency**: sustained neuronal firing outstrips oxygen and glucose supply and causes permanent injury.

Key idea — the postictal state. After a generalised seizure the patient is typically drowsy, confused, and may have a headache or transient focal weakness. This is expected recovery, not a new deficit — but a **prolonged** or **deepening** postictal state needs re-evaluation.

Levels of consciousness

Arousal depends on the **reticular activating system** in the brainstem plus functioning cortex. The **Glasgow Coma Scale** scores **eye opening, verbal response, and motor response**; the lowest possible total is 3 and the highest is 15. **Decorticate** posturing (arms

flexed toward the core) indicates cortical injury; **decerebrate** (arms extended) indicates deeper brainstem injury and is more ominous.

MNEMONIC DeCORTicate = arms to the CORE. Decerebrate is extended — and worse.

PRACTICE

Q1. Which neurotransmitter deficiency causes the motor features of Parkinson disease?

- A. Dopamine, from degeneration of the substantia nigra
- B. Acetylcholine at the neuromuscular junction
- C. Serotonin in the brainstem
- D. GABA in the cerebellum

A is correct. Loss of nigrostriatal dopamine removes the facilitation of movement, producing the resting tremor, rigidity, and bradykinesia triad.

PRACTICE

Q2. Why is status epilepticus a medical emergency?

- A. Sustained neuronal firing outstrips oxygen and glucose supply, causing permanent neuronal injury
- B. It always indicates a brain tumour
- C. It causes immediate cerebral haemorrhage
- D. It is contagious

A is correct. Continuous firing has enormous metabolic demand while the airway and ventilation are compromised, so neurons die and systemic complications follow.

PRACTICE

Q3. Which posturing indicates the deeper, more ominous injury?

- A. Decerebrate – rigid extension of the arms
- B. Decorticate – arms flexed toward the core
- C. Both are equivalent
- D. Neither indicates brain injury

A is correct. Decorticate posturing reflects cortical injury above the brainstem; decerebrate reflects brainstem-level damage and carries a worse prognosis. Progression from decorticate to decerebrate signals deterioration.

WEEK 5

Neuro Assessment, the NMJ & Trauma

CNS infection, the neuromuscular junction, spinal cord injury and its distinctive shock, and the demyelinating disorders.

17 Meningitis & CNS Infection

Key idea — meningeal irritation produces the classic triad: fever, headache, and nuchal rigidity (neck stiffness), usually with photophobia. Inflamed meninges resist stretch, which is what makes neck flexion painful and produces positive Kernig and Brudzinski signs.

CSF finding	BACTERIAL	VIRAL
Appearance	Cloudy / turbid	Clear
White cells	↑↑ Neutrophils	↑ Lymphocytes
Glucose	↓↓ LOW — bacteria consume it	Normal
Protein	↑↑ High	↑ Mildly high
Course	Rapid, can kill in hours — antibiotics immediately	Usually self-limiting, supportive care

MNEMONIC Bacteria eat the sugar. A **LOW CSF glucose** is the single most useful discriminator “it means something is metabolising it.”

RED FLAG — MENINGOCOCCAL DISEASE

Fever and neck stiffness with a **petechial or purpuric rash** that does not blanch suggests *Neisseria meningitidis*. It can progress to septic shock and DIC within hours, so antibiotics are given **without waiting** for confirmation, and close contacts need prophylaxis.

Pearl — encephalitis vs meningitis. **Meningitis** inflames the coverings: fever, headache, stiff neck, but mentation is relatively preserved. **Encephalitis** inflames the brain itself, so **altered mental status, seizures, and focal deficits** dominate.

PRACTICE

Q1. CSF shows cloudy fluid, high neutrophils, high protein, and LOW glucose. Which is it?

- A. Bacterial meningitis
- B. Viral meningitis
- C. Multiple sclerosis
- D. Normal CSF

A is correct. Bacteria metabolise CSF glucose, so a low glucose with neutrophils and turbidity is the bacterial pattern. Viral meningitis leaves glucose normal.

PRACTICE

Q2. Which finding most distinguishes encephalitis from meningitis?

- A. Altered mental status, seizures, and focal neurologic deficits
- B. Fever
- C. Headache
- D. Photophobia

A is correct. Fever, headache, and photophobia occur in both. Involvement of brain parenchyma is what adds altered cognition, seizures, and focal signs.

PRACTICE

Q3. A patient has fever, neck stiffness, and a non-blanching petechial rash. What is the priority?

- A. Immediate antibiotics – meningococcal disease can cause shock and DIC within hours
- B. Await lumbar puncture results before treating
- C. Give antipyretics and reassess in the morning
- D. Begin antiviral therapy

A is correct. The rash suggests meningococcaemia. Because deterioration is measured in hours, treatment precedes confirmation.

18 The Neuromuscular Junction

The NMJ is the synapse where a motor neuron meets muscle. The nerve releases **acetylcholine**, which crosses the cleft and binds receptors on the muscle to trigger contraction. Disorders here cause **weakness with normal sensation**.

Disorder	Lesion	Signature
Myasthenia gravis	Autoantibodies block/destroy ACh receptors on the muscle side	Weakness WORSENS with use, improves with rest. Ptosis and diplopia first; worse at end of day; may progress to swallowing and breathing
Botulism	Toxin blocks ACh release from the nerve	Descending flaccid paralysis, starting with cranial nerves — blurred vision, dysphagia — then respiratory muscles
Organophosphate poisoning	Inhibits acetylcholinesterase → too much ACh	SLUDGE : salivation, lacrimation, urination, defecation, GI upset, emesis; plus miosis and bradycardia
Guillain-Barré© syndrome	Autoimmune demyelination of peripheral nerves (not strictly NMJ)	ASCENDING weakness starting in the legs, areflexia , often after an infection

Key idea — direction tells you the diagnosis. Guillain-Barré© ascends (feet upward). **Botulism descends** (cranial nerves downward). Both threaten the **diaphragm**, which is what actually kills — so monitoring respiratory function is the priority in either.

MNEMONIC Myasthenia = Muscles tire. Use makes it worse, rest makes it better. Contrast with muscular dystrophy, where weakness is constant and progressive regardless of activity.

RED FLAG — MYASTHENIC CRISIS

When weakness spreads to the muscles of **swallowing and breathing**, the patient cannot clear secretions or ventilate. Difficulty handling saliva, a weak cough, or a falling vital capacity means ventilatory support may be needed — do not wait for oxygen saturation to fall, because it drops late.

PRACTICE

Q1. Which pattern is characteristic of myasthenia gravis?

- A. Weakness that worsens with sustained activity and improves after rest
- B. Constant weakness beginning in the calves and progressing steadily
- C. Ascending weakness with loss of reflexes after an infection
- D. Descending paralysis starting with the cranial nerves

A is correct. Fatigability localises the problem to the neuromuscular junction. Option C is Guillain-Barré and option D is botulism.

PRACTICE

Q2. What is the priority assessment in Guillain-Barré syndrome?

- A. Respiratory function, because ascending paralysis can reach the diaphragm
- B. Level of consciousness
- C. Pupillary response
- D. Blood glucose

A is correct. The weakness climbs from the legs upward, and respiratory muscle involvement is what makes it lethal. Ventilatory monitoring drives management.

PRACTICE

Q3. Organophosphate poisoning produces excessive salivation, lacrimation, and bradycardia. Why?

- A. Acetylcholinesterase is inhibited, so acetylcholine accumulates and overstimulates receptors
- B. Acetylcholine release is blocked at the nerve terminal
- C. Acetylcholine receptors are destroyed by antibodies
- D. Peripheral nerves are demyelinated

A is correct. Without the enzyme that breaks it down, acetylcholine persists in the synapse and drives continuous parasympathetic activity – the SLUDGE picture.

19 Spinal Cord Injury & Neurogenic Shock

Key idea — level determines function. Injury above **C3–C5** threatens the diaphragm ("C3, 4, 5 keep the diaphragm alive"). **Cervical** injury → **tetraplegia**; **thoracic or lumbar** → **paraplegia**. **Complete** injury = total loss below the level; **incomplete** = some function preserved.

	SPINAL SHOCK	NEUROGENIC SHOCK
What it is	A neurologic phenomenon — temporary loss of all reflex activity below the injury	A circulatory phenomenon — a true shock state from lost sympathetic tone
Findings	Flaccid paralysis, absent reflexes, no sensation; resolves over days to weeks	Hypotension with BRADYCARDIA , warm dry flushed skin
Typical level	Any level	Usually T6 or above

Pearl — the bradycardia is the tell. Every other shock state produces **tachycardia with cool, clammy skin**. Neurogenic shock produces **bradycardia with warm, dry skin**, because the sympathetic system that would normally speed the heart and constrict vessels has been disconnected.

RED FLAG — AUTONOMIC DYSREFLEXIA

In injuries at **T6 or above**, a noxious stimulus below the injury (most often a **blocked catheter or full bladder**, or constipation) triggers massive unopposed sympathetic discharge: **severe hypertension, pounding headache, flushing and sweating above the injury, pallor below, and bradycardia**. It can cause stroke. Management is to **sit the patient upright and remove the trigger** — usually relieving the bladder.

PRACTICE

Q1. Which finding distinguishes neurogenic shock from hypovolaemic shock?

- A. Bradycardia with warm, dry skin
- B. Hypotension
- C. Reduced urine output
- D. Altered mental status

A is correct. All shock states share hypotension and poor perfusion. Only neurogenic shock loses sympathetic tone, so the heart cannot speed up and the skin stays warm and dry.

PRACTICE

Q2. A patient with a T4 injury develops a pounding headache, flushing above the lesion, and blood pressure of 210/110. What is the first action?

- A. Sit the patient upright and look for the trigger, usually a blocked catheter or full bladder
- B. Lay the patient flat and elevate the legs
- C. Administer a fluid bolus
- D. Apply warming blankets

A is correct. This is autonomic dysreflexia. Sitting up uses gravity to lower blood pressure while the noxious stimulus is found and removed “laying the patient flat would worsen it.

PRACTICE

Q3. Why does an injury at C4 threaten life immediately?

- A. The phrenic nerve arises from C3 to C5 and innervates the diaphragm
- B. The cardiac conduction system originates at C4
- C. C4 controls the swallowing reflex
- D. The vagus nerve exits at C4

A is correct. Losing phrenic innervation paralyses the diaphragm, so ventilation fails. This is why cervical injuries are managed with immediate attention to the airway and breathing.

20 Demyelination & Trauma Overview

	Multiple sclerosis	Guillain-Barré syndrome
Where	CNS — oligodendrocytes	PNS — Schwann cells
Course	Chronic , relapsing-remitting or progressive	Acute , monophasic, usually after an infection
Pattern	Lesions separated in time and space — optic neuritis, sensory change, weakness, ataxia	Symmetric ASCENDING weakness with areflexia
Recovery	Incomplete; accumulates disability	Usually good once remyelination occurs

Pearl — heat and MS. Raised body temperature further slows conduction in already-demyelinated axons, so symptoms transiently worsen with hot weather, hot baths, or fever. This is a conduction effect, not a true relapse.

Burns — the trauma overlap

Depth	Appearance	Pain
Superficial	Red, dry, no blisters	Painful
Partial thickness	Blisters , moist, weeping	Very painful
Full thickness	White, leathery, dry	Painless — nerve endings destroyed

Key idea — the two phases of burn danger. First 24–48 hours: fluid. Massive capillary leak shifts plasma into tissue → **hypovolaemic shock**, haemoconcentration, and **hyperkalaemia** from cell lysis. **After that: infection**, because the barrier is gone and necrotic tissue is a culture medium. Also watch **hypothermia** and, with facial burns, **airway oedema**.

Pearl — the painless burn is the worst burn. Absence of pain in a leathery white burn means the full thickness of dermis, including nerve endings, has been destroyed.

PRACTICE

Q1. Which feature separates multiple sclerosis from Guillain-Barré syndrome?

- A. MS is a chronic CNS demyelination with lesions separated in time and space; GBS is an acute PNS demyelination
- B. MS affects peripheral nerves and GBS affects the brain
- C. Both are acute and monophasic
- D. Neither involves myelin

A is correct. The location (CNS vs PNS) and the course (chronic relapsing vs acute monophasic) are the two clean discriminators.

PRACTICE

Q2. During the first 24 hours after a major burn, which problem is the greatest threat?

- A. Hypovolaemic shock from massive capillary leak
- B. Wound infection and sepsis
- C. Contracture formation
- D. Hypokalaemia

A is correct. Plasma shifts into the interstitium, causing burn shock. Infection dominates after 48 hours, and early cell lysis causes HYPERkalaemia rather than hypokalaemia.

PRACTICE

Q3. A burn is white and leathery, and the patient reports no pain in that area. What does this indicate?

- A. Full-thickness burn with destruction of dermal nerve endings
- B. A superficial burn that is healing
- C. Effective analgesia
- D. A partial-thickness burn

A is correct. Pain requires intact nerve endings. Their destruction — with the leathery white appearance — indicates the deepest injury, which will need grafting.

WEEK 6

The Endocrine System

Feedback loops and what happens at each end of them — the unit with the most lab-signature questions on the exam.

21 Hormones & Feedback

Hypothalamus (releasing hormone) → **Anterior pituitary** (stimulating hormone) → **Target gland** (final hormone) → the final hormone **feeds back** to switch off both levels above it.

Key idea — negative feedback is the whole logic of this unit. The final hormone shuts off its own upstream signals. So to locate a lesion, compare the stimulating hormone with the final hormone:

- **Final hormone HIGH + stimulating hormone LOW** → the problem is the **target gland** (primary) — it is overproducing autonomously.
- **Final hormone LOW + stimulating hormone HIGH** → the problem is the **target gland** (primary) — it is failing and the pituitary is shouting.
- **BOTH low** → the problem is the **pituitary or hypothalamus** (secondary/tertiary).

MNEMONIC They move **OPPOSITE** in primary disease, **TOGETHER** in secondary. One line answers most endocrine lab questions on the exam.

Class	Property	Examples
Peptide/protein	Water soluble; bind surface receptors; act fast	Insulin, ADH, growth hormone, PTH
Steroid	Lipid soluble; cross the membrane to intracellular receptors; slower, longer acting	Cortisol, aldosterone, oestrogen, testosterone
Amine	Mixed properties	Thyroid hormone, catecholamines

Pearl — up-regulation and down-regulation. Chronic **excess** hormone causes target cells to reduce receptor numbers (**down-regulation**) — a mechanism behind insulin resistance. Chronic **deficiency** increases receptors (**up-regulation**), making the tissue more sensitive.

PRACTICE

Q1. Labs show a HIGH final hormone with a LOW stimulating hormone. Where is the lesion?

- A. The target gland " it is overproducing and suppressing the pituitary
- B. The anterior pituitary
- C. The hypothalamus
- D. The feedback receptors only

A is correct. An autonomously overactive gland raises the final hormone, and normal negative feedback then suppresses the stimulating hormone. Opposite movement means primary gland disease.

PRACTICE

Q2. Both the stimulating hormone AND the final hormone are low. What does this indicate?

- A. Secondary failure " the pituitary or hypothalamus is not signalling
- B. Primary target gland failure
- C. Primary target gland excess
- D. Normal feedback

A is correct. If the gland were the problem, the pituitary would compensate by raising its stimulating hormone. Both being low means the drive itself has failed.

PRACTICE

Q3. Why does chronic hormone excess reduce target tissue responsiveness?

- A. Down-regulation â€” target cells decrease their receptor numbers
- B. The hormone is destroyed faster
- C. The gland stops producing the hormone
- D. Feedback becomes positive instead of negative

A is correct. Persistent stimulation causes cells to withdraw receptors from the surface, blunting the response. This is one contributor to insulin resistance in type 2 diabetes.

	SIADH (too much ADH)	DIABETES INSIPIDUS (too little ADH)
Water handling	Retains water	Loses water
Serum sodium	↓ Low (dilutional)	↑ High (concentrated)
Serum osmolality	↓ Low	↑ High
Urine	Concentrated , scant	Dilute , enormous volumes (polyuria)
Weight / volume	Gain, fluid overload	Loss, dehydration, intense thirst
Danger	Cerebral oedema and seizures from hyponatraemia	Hypovolaemic shock if the patient cannot drink
Causes	Small cell lung cancer (paraneoplastic), CNS disorders, some drugs	Head injury, pituitary surgery, tumour; or renal resistance to ADH

MNEMONIC SIADH = **Soaked Inside** (water retained, sodium diluted). DI = **Dry Inside** (water pouring out, sodium concentrated). They are mirror images at every line of the table.

Key idea — the urine is the giveaway. Both present with abnormal sodium, but ask what the urine is doing. **Inappropriately concentrated urine with low serum sodium** = SIADH. **Massive dilute urine with high serum sodium** = DI.

Other pituitary disorders	Feature
Growth hormone excess	Before growth plates close → gigantism ; after → acromegaly (enlarging hands, feet, jaw, coarse features)
Growth hormone deficiency	Short stature with normal proportions in children
Prolactinoma	Galactorrhoea, amenorrhoea, infertility; may compress the optic chiasm → visual field loss

PRACTICE

Q1. A patient has serum sodium 122, low serum osmolality, and concentrated urine. Which condition is this?

- A. SIADH – excess ADH causes water retention and dilutional hyponatraemia
- B. Diabetes insipidus
- C. Dehydration
- D. Diabetic ketoacidosis

A is correct. Retaining water dilutes the serum while the kidney, under ADH influence, continues concentrating urine – an inappropriate combination that defines SIADH.

PRACTICE

Q2. After pituitary surgery a patient produces 6 litres of dilute urine in a day and is intensely thirsty. What is happening?

- A. Diabetes insipidus from insufficient ADH
- B. SIADH
- C. Normal postoperative diuresis
- D. Acute kidney injury

A is correct. ADH is released from the posterior pituitary, so surgery there can abolish it. Without ADH the collecting ducts cannot reabsorb water and huge volumes of dilute urine are lost.

PRACTICE

Q3. Growth hormone excess in an adult produces which condition?

- A. Acromegaly – enlargement of hands, feet, and facial bones
- B. Gigantism
- C. Dwarfism
- D. Cushing syndrome

A is correct. Once the epiphyseal plates have fused, bones can no longer lengthen, so they thicken instead. Gigantism occurs only if the excess begins before closure.

Key idea — thyroid hormone sets the metabolic rate. Every symptom follows from that single fact. **Too much** = everything speeds up. **Too little** = everything slows down. You do not need to memorise two symptom lists, only the direction.

	HYPERthyroid (Graves)	HYPOthyroid (Hashimoto)
Labs	↓ TSH, ↑ free T4	↑ TSH, ↓ free T4
Mechanism	Autoantibodies STIMULATE the TSH receptor (type II hypersensitivity)	Autoimmune DESTRUCTION of thyroid tissue
Weight / appetite	Loss despite increased appetite	Gain despite poor appetite
Temperature	Heat intolerance, sweating	Cold intolerance
Cardiac	Tachycardia, palpitations, atrial fibrillation, hypertension	Bradycardia, low output
Neuro / mood	Anxiety, tremor, insomnia, hyperreflexia	Fatigue, depression, slowed thinking, delayed reflexes
GI	Diarrhoea	Constipation
Distinctive	Exophthalmos (bulging eyes), goitre, warm moist skin	Dry coarse skin, hair loss, non-pitting myxoedema , hoarse voice

Pearl — why exophthalmos happens in Graves and does not resolve with treatment. The same antibodies stimulate retro-orbital tissue to swell, pushing the eyes forward. Because it is a tissue change rather than a hormone effect, normalising thyroid levels does not reverse it.

RED FLAG — THE TWO THYROID EMERGENCIES

Thyroid storm: extreme hypermetabolism — high fever, severe tachycardia or atrial fibrillation, agitation progressing to delirium, vomiting. Often triggered by infection, surgery, or stress.

Myxoedema coma: the opposite — **hypothermia, bradycardia, hypoventilation, and declining consciousness.** Both are frequently fatal untreated.

CROSS-UNIT THREAD Graves and Hashimoto are both **autoimmune** (Week 2). Graves is a **type II** hypersensitivity where the antibody *stimulates* rather than destroys — an unusual and frequently tested variation.

PRACTICE

Q1. Labs show TSH 0.02 (low) with a high free T4. Which condition is this?

- A. Primary hyperthyroidism, such as Graves disease
- B. Primary hypothyroidism
- C. Secondary hypothyroidism
- D. Euthyroid sick syndrome

A is correct. The gland is overproducing, so negative feedback suppresses TSH. Values moving in opposite directions indicates primary gland disease.

PRACTICE

Q2. Which set of findings fits hypothyroidism?

- A. Weight gain, cold intolerance, constipation, bradycardia, delayed reflexes
- B. Weight loss, heat intolerance, diarrhoea, tachycardia
- C. Weight loss, cold intolerance, constipation
- D. Weight gain, heat intolerance, diarrhoea

A is correct. Everything slows when metabolic rate falls. If you can recall the direction, you can derive the entire symptom list rather than memorising it.

PRACTICE

Q3. A hyperthyroid patient develops fever of 40°C, heart rate 160, and agitation after surgery. What is this?

- A. Thyroid storm – a life-threatening hypermetabolic crisis
- B. Myxoedema coma
- C. A routine postoperative fever
- D. Addisonian crisis

A is correct. Surgery, infection, and trauma are classic triggers. Uncontrolled hypermetabolism causes hyperthermia and high-output cardiac failure and demands immediate treatment.

24 Adrenal

The adrenal **cortex** makes three steroid classes — **aldosterone** (salt), **cortisol** (sugar), and androgens (sex). The **medulla** makes catecholamines.

MNEMONIC Salt, Sugar, Sex — the deeper you go, the sweeter it gets. Zona glomerulosa (aldosterone) → fasciculata (cortisol) → reticularis (androgens).

	CUSHING (cortisol excess)	ADDISON (cortisol deficiency)
Body habitus	Truncal obesity , moon face, buffalo hump, thin extremities	Weight loss , muscle wasting, fatigue
Skin	Thin, easy bruising, purple striae , poor healing	Hyperpigmentation (from high ACTH), especially creases and scars
Glucose	↑ Hyperglycaemia — cortisol is a stress hormone that raises glucose	↓ Hypoglycaemia
Sodium / potassium	↑ Na ⁺ , ↓ K ⁺ ; hypertension, fluid retention	↓ Na ⁺ , ↑ K ⁺ ; hypotension, salt craving
Immune	Suppressed — infection risk, poor healing	—
Common cause	Exogenous steroids (most common overall), pituitary adenoma, adrenal tumour	Autoimmune destruction; abrupt steroid withdrawal

Key idea — every Cushing sign traces to cortisol's normal jobs, exaggerated. Cortisol raises glucose (→ hyperglycaemia), breaks down protein (→ thin skin, striae, muscle wasting, poor healing), redistributes fat centrally (→ moon face, buffalo hump), suppresses immunity (→ infection), and has mild mineralocorticoid effect (→ sodium retention, hypertension).

RED FLAG — ADDISONIAN CRISIS

Cortisol enables the stress response. A patient who **stops long-term steroids abruptly**, or who has Addison disease and meets a stressor (infection, surgery, trauma), cannot mount that response → **hypotension, vomiting, hyponatraemia, hyperkalaemia, and shock**. This is why steroids are always tapered rather than stopped.

Pearl — why Addison causes hyperpigmentation. The failing adrenal cannot make cortisol, so ACTH rises to drive it. ACTH shares a precursor with melanocyte-stimulating hormone, so very high levels also darken the skin. A tan that appears in creases and old scars is the clue.

Phaeochromocytoma — a catecholamine-secreting medullary tumour. Classic **episodic** triad: **headache, palpitations, and sweating**, with severe paroxysmal hypertension.

PRACTICE

Q1. Which laboratory pattern fits Addison disease?

- A. Low sodium, high potassium, low glucose, low cortisol with high ACTH
- B. High sodium, low potassium, high glucose, high cortisol
- C. High sodium, high potassium, high cortisol
- D. Low sodium, low potassium, low ACTH

A is correct. Losing aldosterone wastes sodium and retains potassium; losing cortisol drops glucose; and the failing gland drives ACTH up, which also causes hyperpigmentation.

PRACTICE

Q2. Why do patients with Cushing syndrome bruise easily and heal poorly?

- A. Cortisol breaks down protein, thinning skin and connective tissue and impairing collagen synthesis
- B. Cortisol destroys platelets
- C. Cortisol causes vitamin K deficiency
- D. Cortisol increases capillary permeability

A is correct. Cortisol is catabolic. Protein breakdown thins the skin and its supporting vessels, producing fragility, purple striae, and delayed healing — the same reason steroids impair wound healing generally.

PRACTICE

Q3. A patient on long-term prednisone stops it abruptly and becomes hypotensive and vomiting. What has happened?

- A. Addisonian crisis – the suppressed adrenals cannot produce cortisol for the stress response
- B. Thyroid storm
- C. SIADH
- D. Cushing syndrome

A is correct. Chronic exogenous steroids suppress the axis by negative feedback and the glands atrophy. Stopping abruptly leaves no cortisol at all, producing acute adrenal insufficiency and shock.

25 Diabetes Mellitus

	TYPE 1	TYPE 2
Defect	Autoimmune destruction of beta cells → absolute insulin deficiency	Insulin resistance plus a relative deficiency
Onset	Usually childhood or adolescence; abrupt	Usually adult; gradual, often asymptomatic for years
Body habitus	Often thin, with weight loss at diagnosis	Commonly overweight, central obesity
Ketosis	Prone — no insulin at all to suppress fat breakdown	Resistant — residual insulin blocks ketogenesis
Treatment	Insulin always	Lifestyle, oral agents, sometimes insulin

Key idea — the classic symptoms all follow from glucose in the urine. Glucose exceeds the renal threshold and spills into urine, dragging water osmotically → **polyuria** → dehydration → **polydipsia**. Cells cannot access glucose despite plenty in the blood → **polyphagia** and **weight loss**. Three Ps from one mechanism.

	DKA	HHS
Typical type	Type 1	Type 2
Glucose	High (often 300–600)	Extremely high (often >600)
Ketones / pH	Present; pH LOW (acidosis)	Absent or minimal; pH normal
Hallmark	Kussmaul respirations , fruity breath, abdominal pain	Profound dehydration and altered mental status
Onset	Hours to a day	Days

MNEMONIC DKA has the **K** for **Ketones** and **Kussmaul**. HHS is **H**igher glucose, **H**yperosmolar, but no ketones — because a little insulin is still present.

Pearl — the potassium trap in DKA. Acidosis drives potassium out of cells, so the **serum** potassium may be normal or high even though **total body** potassium is severely depleted. As insulin and fluids correct the acidosis, potassium moves back into cells and can plummet — which is why potassium is monitored closely during treatment.

Key idea — long-term complications split in two. Microvascular (small vessels): **retinopathy, nephropathy, neuropathy**. **Macrovascular** (large vessels): **coronary artery disease, stroke, peripheral arterial disease**. Add impaired leukocyte function and neuropathy, and you have the diabetic foot ulcer that will not heal.

RED FLAG — HYPOGLYCAEMIA

Faster and more immediately dangerous than hyperglycaemia. Sweating, shakiness, tachycardia, hunger, then confusion, seizure, and coma. **The brain has no glucose reserve**, so treatment is immediate.

PRACTICE

Q1. Why does DKA occur in type 1 rather than type 2 diabetes?

- A. Type 1 has an absolute insulin deficiency, so fat breakdown proceeds unchecked and generates ketoacids
- B. Type 1 patients have higher blood glucose
- C. Type 2 patients do not experience dehydration
- D. Type 1 involves insulin resistance

A is correct. Even a small amount of insulin suppresses ketogenesis. In type 2 that residual insulin prevents ketosis, so extreme hyperglycaemia produces HHS instead.

PRACTICE

Q2. Why does a patient in DKA breathe deeply and rapidly?

- A. Kussmaul respirations "the lungs blow off CO₂, to compensate for metabolic acidosis
- B. Ketones irritate the airway directly
- C. Dehydration stimulates the respiratory centre
- D. It is a primary respiratory alkalosis

A is correct. Ketoacids consume bicarbonate and drop the pH. The medulla responds by driving ventilation to eliminate CO₂, "respiratory compensation for a metabolic problem, straight from Week 1.

PRACTICE

Q3. Why must potassium be watched closely while treating DKA?

- A. Correcting the acidosis with insulin shifts potassium back into cells, unmasking a total-body deficit
- B. Insulin causes the kidneys to retain potassium
- C. Potassium rises steadily throughout treatment
- D. Potassium is unaffected by insulin

A is correct. The initial serum potassium looks reassuring because acidosis pushed it out of cells. Treatment reverses that shift and can drop it dangerously fast.

PUT IT TOGETHER

Cross-Unit Threads & Confusion Pairs

The links that reward students who see the whole course, and the look-alikes that cost the most marks.

1. ATP failure is the common path of cell injury

W1 → W4 Whether the cause is hypoxia, ischemia, or a toxin, the sequence is the same: ATP falls → the sodium pump fails → the cell swells → membranes rupture. It is why an ischemic stroke (Week 4) and a hypoxic cell in a test tube (Week 1) die the same way.

2. Chronic inflammation drives cancer

W2 → W3 Sustained injury means sustained repair, and every cell division is a chance for an uncorrected mutation. Reflux → Barrett → adenocarcinoma; *H. pylori* → gastritis → gastric cancer; hepatitis → cirrhosis → hepatocellular carcinoma.

3. Metaplasia → dysplasia → carcinoma

W1 → W3 The adaptation vocabulary from Week 1 is the carcinogenesis sequence from Week 3. Metaplasia is adaptive and reversible; dysplasia is disordered and premalignant; carcinoma in situ becomes invasive when it breaches the basement membrane.

4. Autoimmunity is one mechanism with many faces

W2 → W5 · W6 Loss of self-tolerance produces **myasthenia gravis** (ACh receptors, Week 5), **multiple sclerosis** (CNS myelin, Week 5), **Graves and Hashimoto** (thyroid, Week 6), **type 1 diabetes** (beta cells, Week 6), and **Addison** (adrenal cortex). One Week 2 concept explains five later diseases.

5. The rigid skull turns any volume into pressure

W1 → W4 Oedema is oedema — but inside a fixed box it becomes a perfusion emergency. Because CO₂ is a cerebral vasodilator, the Week 1 acid–base concept directly worsens the Week 4 intracranial pressure cascade.

6. Every feedback loop breaks the same three ways

W6 THROUGHOUT Gland overactive (final hormone high, stimulating hormone low) · gland failing (final low, stimulating high) · pituitary failing (both low). Thyroid, adrenal, and gonadal axes all read the same way once you know the rule.

7. Acidosis and potassium move together

W1 → W6 Acidosis pushes potassium out of cells, so DKA presents with a normal or high serum potassium despite total-body depletion. Correct the acidosis and potassium plunges. A Week 1 electrolyte principle governing a Week 6 emergency.

8. Shock is one endpoint reached by many roads

W2 · W5 · W6 Anaphylaxis (Week 2, distributive), **neurogenic shock** (Week 5, lost sympathetic tone), **addisonian crisis** (Week 6, no cortisol for the stress response), and **DKA** (Week 6, osmotic dehydration). All end in inadequate perfusion → anaerobic metabolism → lactic acidosis.

PRACTICE

Q1. Which mechanism links chronic gastro-oesophageal reflux to oesophageal adenocarcinoma?

- A. Chronic injury drives metaplasia, then dysplasia, then carcinoma
- B. Acid directly mutates DNA in a single exposure
- C. Reflux causes immunodeficiency
- D. Reflux blocks lymphatic drainage

A is correct. Repeated injury and repair means repeated cell division. Metaplastic tissue is more acid-resistant but abnormal for that site, and sustained long enough it progresses toward malignancy.

PRACTICE

Q2. A patient in DKA has a serum potassium of 5.4. What should be anticipated during treatment?

- A. Potassium will fall as insulin and fluids correct the acidosis and drive it back into cells
- B. Potassium will continue to rise
- C. Potassium will remain unchanged
- D. Potassium is irrelevant in DKA

A is correct. The elevated reading reflects a transcellular shift, not a surplus. Total body potassium is depleted, and correcting the acidosis can produce dangerous hypokalaemia.

PRACTICE

Q3. Anaphylaxis, neurogenic shock, and addisonian crisis share which final common pathway?

- A. Inadequate tissue perfusion leading to anaerobic metabolism and lactic acidosis
- B. Antibody-mediated destruction of red cells
- C. Raised intracranial pressure
- D. Loss of self-tolerance

A is correct. The routes differ – mast cell mediators, lost sympathetic tone, absent cortisol – but all end with cells that cannot get enough oxygen, which is what defines shock.

27 Confusion Pairs — A vs B

Most missed questions are not gaps in knowledge — they are two things getting mixed up. Here is the single discriminator for each.

Pair	The one clue that separates them
Hypertrophy vs hyperplasia	Bigger cells vs more cells
Metaplasia vs dysplasia	Orderly substitution (reversible) vs disordered, atypical (pre-malignant)
Necrosis vs apoptosis	Inflammation present → necrosis
Autosomal dominant vs recessive	Every generation vs skips generations (silent carriers)
Respiratory vs metabolic acidosis	CO ₂ moves opposite to pH → respiratory (ROME)
Vomiting vs diarrhoea	Loses acid → alkalosis ; loses bicarbonate → acidosis
Innate vs adaptive immunity	Memory → adaptive
Active vs passive immunity	You made it (lasting) vs you received it (no memory)
Type I vs type IV hypersensitivity	Minutes and IgE vs 24–72 hours and T cells
Type II vs type III	Antibody against a specific tissue vs complexes depositing anywhere
Acute vs chronic inflammation	Neutrophils vs macrophages and lymphocytes (with fibrosis)
Benign vs malignant	Invasion and metastasis — nothing else is definitive
Oncogene vs tumour suppressor	Stuck accelerator (one hit) vs failed brake (both copies)
Staging vs grading	Spread vs appearance down the microscope
Carcinoma vs sarcoma	Epithelium vs connective tissue
Ischemic vs hemorrhagic stroke	Occlusion vs rupture — severe sudden headache favours haemorrhage
TIA vs stroke	Full resolution within 24 h with no infarction
Epidural vs subdural haematoma	Arterial with a lucid interval vs venous, slower, older adults

Decorticate vs decerebrate	Arms to the core vs extended — extension is worse
Cushing triad vs shock	BP up with pulse down vs BP down with pulse up
Broca vs Wernicke aphasia	Broken speech (aware) vs word salad (unaware)
Bacterial vs viral meningitis	LOW CSF glucose → bacterial
Meningitis vs encephalitis	Altered mentation, seizures, focal signs → encephalitis
Guillain-Barré vs botulism	Ascending vs descending paralysis
Myasthenia vs muscular dystrophy	Fatigable (worse with use) vs constant and progressive
Spinal vs neurogenic shock	Neurologic areflexia vs a circulatory state with hypotension and bradycardia
Neurogenic vs hypovolaemic shock	Bradycardia, warm dry skin vs tachycardia, cool clammy skin
MS vs Guillain-Barré	CNS chronic relapsing vs PNS acute monophasic
SIADH vs diabetes insipidus	Concentrated urine with low sodium vs dilute urine with high sodium
Graves vs Hashimoto	Low TSH, high T4 vs high TSH, low T4
Thyroid storm vs myxoedema coma	Fever and tachycardia vs hypothermia and bradycardia
Cushing vs Addison	High glucose, high sodium, low potassium vs low glucose, low sodium, high potassium
Type 1 vs type 2 diabetes	Absolute deficiency, ketosis-prone vs resistance , ketosis-resistant
DKA vs HHS	Ketones and low pH vs no ketones, normal pH, higher osmolality

ABG quick table

pH	PaCO ₂	HCO ₃	Interpretation
↓	↑	n	Uncompensated respiratory acidosis (acute)
n (<7.40)	↑	↑	Fully compensated respiratory acidosis (chronic)
↑	↓	n	Uncompensated respiratory alkalosis (anxiety, pain, fever)
↓	n	↓	Uncompensated metabolic acidosis (DKA, shock, diarrhoea)
↓	↓	↓	Partially compensated metabolic acidosis (Kussmaul)
↑	n	↑	Uncompensated metabolic alkalosis (vomiting, NG suction)

REMEMBER ROME — Respiratory Opposite, Metabolic Equal. The value matching the pH direction is the problem; the one opposing it is the compensation.

Endocrine lab signatures

Condition	Pattern	Clinical giveaway
Graves (hyperthyroid)	↓TSH, ↑T4	Weight loss, heat intolerance, exophthalmos
Hashimoto (hypothyroid)	↑TSH, ↓T4	Weight gain, cold intolerance, delayed reflexes
Secondary hypothyroid	↓TSH, ↓T4	Both low → pituitary problem
Cushing	↑cortisol, ↑glucose, ↑Na, ↓K	Moon face, striae, truncal obesity
Addison	↓cortisol, ↑ACTH, ↓Na, ↑K, ↓glucose	Hyperpigmentation, salt craving, hypotension
SIADH	↓Na, ↓serum osm, ↑urine osm	Fluid retention, confusion, seizures
Diabetes insipidus	↑Na, ↑serum osm, ↓urine osm	Polyuria, intense thirst, dehydration
DKA	↑glucose, ↑ketones, ↓pH, ↓HCO ₃	Kussmaul breathing, fruity breath
HHS	↑↑glucose, ↑osm, no ketones , normal pH	Profound dehydration, altered mentation

Don't-miss emergencies (Weeks 1–6)

Emergency	The tell	Why it cannot wait
Severe hyperkalaemia	K^+ >6 with peaked T waves	Ventricular fibrillation and arrest
Severe hyponatraemia	Na^+ <120 with seizures	Cerebral oedema
Anaphylaxis	Facial swelling, stridor, hypotension after exposure	Airway obstruction and distributive shock — epinephrine now
Neutropenic fever	Fever in a chemotherapy patient	Fever may be the only sign of overwhelming infection
Spinal cord compression	Back pain + leg weakness + retention in a cancer patient	Permanent paralysis
Tumour lysis syndrome	$\uparrow K$, $\uparrow$ phosphate, $\uparrow$ uric acid, $\downarrow$ calcium after chemotherapy	Dysrhythmias and acute kidney injury
Rising ICP	Falling level of consciousness ; late = Cushing triad	Herniation
Blown pupil	New unilateral fixed dilated pupil	Third nerve compression — herniation in progress
Acute stroke	Sudden focal deficit within the treatment window	The penumbra is dying — time is brain
Epidural haematoma	Lucid interval then rapid decline	Arterial bleed filling a closed space
Status epilepticus	Seizure not stopping or recurring without recovery	Neuronal death and systemic collapse
Bacterial meningitis	Fever + neck stiffness + petechial rash	Death within hours; treat before confirming
Ascending paralysis	Guillain-Barré reaching the chest	Ventilatory failure
Myasthenic crisis	Weak cough, trouble with secretions	Airway and ventilation

Autonomic dysreflexia	T6 or above + severe hypertension + pounding headache	Stroke — sit upright, relieve the trigger
Thyroid storm	Fever, severe tachycardia, agitation	High-output failure and hyperthermia
Myxoedema coma	Hypothermia, bradycardia, declining consciousness	Respiratory and cardiovascular collapse
Addisonian crisis	Hypotension, vomiting, ↓Na, ↑K after stopping steroids	No stress response → shock
DKA	High glucose, ketones, Kussmaul breathing	Acidosis, dehydration, electrolyte collapse
Hypoglycaemia	Sweaty, shaky, confused	The brain has no glucose reserve — seizure and coma in minutes



REHEARSAL

30-Question Cumulative Practice Exam

Mixed order across Weeks 1–6 — the way the real midterm will feel. Answer all 30 before revealing anything.

Q1 · W1

Which change is the earliest sign of reversible hypoxic cell injury?

- A. Cellular swelling
- B. Nuclear fragmentation
- C. Membrane rupture
- D. Calcification

A. ATP failure stops the sodium pump, so water follows sodium into the cell. Swelling is early and reversible; the others indicate irreversible injury.

Q2 · W6

TSH is high and free T4 is low. Which condition is this?

- A. Primary hypothyroidism
- B. Graves disease
- C. Secondary hypothyroidism
- D. Thyroid storm

A. The gland is failing and the pituitary compensates by raising TSH. Values moving in opposite directions indicates primary gland disease.

Q3 · W2

A rash appears 48 hours after contact with nickel. Which hypersensitivity type?

- A. Type IV (delayed, T-cell mediated)
- B. Type I
- C. Type II
- D. Type III

A. A 24-to-72-hour delay with no antibody involvement is the signature of a T-cell mediated reaction.

Q4 · W4

Which is the earliest indicator of increased intracranial pressure?

- A. A change in level of consciousness
- B. Bradycardia
- C. A fixed dilated pupil
- D. Widening pulse pressure

A. The others are late. Altered consciousness appears while there is still time to intervene.

Q5 · W3

Which feature defines a tumour as malignant?

- A. The capacity to invade and metastasise
- B. Rapid growth
- C. Large size
- D. Causing symptoms

A. Benign tumours can be large, fast-growing, and dangerous by location, but never invade or metastasise.

Q6 · W5

CSF is cloudy with high neutrophils, high protein, and LOW glucose. Which is it?

- A. Bacterial meningitis
- B. Viral meningitis
- C. Multiple sclerosis
- D. Normal CSF

A. Bacteria consume CSF glucose. A low glucose is the single most useful discriminator.

Q7 · W1

ABG: pH 7.50, PaCO₂ 28, HCO₃ 24. Which disorder?

- A. Uncompensated respiratory alkalosis
- B. Metabolic alkalosis
- C. Respiratory acidosis
- D. Compensated metabolic acidosis

A. High pH with a low CO₂, moving opposite to it = respiratory. A normal bicarbonate means no compensation yet.

Q8 · W6

Which laboratory picture fits Addison disease?

- A. Low sodium, high potassium, low glucose
- B. High sodium, low potassium, high glucose
- C. High sodium, high potassium, high glucose
- D. Low sodium, low potassium, low glucose

A. Loss of aldosterone wastes sodium and retains potassium; loss of cortisol drops glucose. Cushing shows the mirror image.

Q9 · W2

Which immunoglobulin crosses the placenta?

- A. IgG
- B. IgM
- C. IgA
- D. IgE

A. IgG alone crosses the placenta, giving the newborn passive protection. IgA comes through breast milk.

Q10 · W5

Hypotension with bradycardia and warm dry skin after a cervical injury indicates:

- A. Neurogenic shock
- B. Hypovolaemic shock
- C. Septic shock
- D. Cardiogenic shock

A. Lost sympathetic tone means the heart cannot speed up and vessels cannot constrict. Every other shock type gives tachycardia and cool skin.

Q11 · W1

Two unaffected parents have a child with an autosomal recessive disease. Recurrence risk per pregnancy?

- A. 25%
- B. 50%
- C. 75%
- D. 100%

A. Both parents are carriers, so each pregnancy independently carries a one-in-four chance of inheriting both recessive alleles.

Q12 · W6

Low serum sodium with concentrated urine and fluid retention indicates:

- A. SIADH
- B. Diabetes insipidus
- C. Dehydration
- D. Addison disease

A. Excess ADH retains water, diluting serum sodium while urine stays inappropriately concentrated. DI is the mirror image.

Q13 · W3

Which best describes the difference between staging and grading?

- A. Staging measures spread; grading measures cellular differentiation
- B. Staging measures differentiation; grading measures spread
- C. Both measure tumour size
- D. Both are microscopic assessments

A. Stage = Spread (TNM). Grade = Glance down a microscope. Staging drives prognosis and treatment.

Q14 · W4

A patient with left-sided weakness most likely has a lesion where?

- A. Right cerebral hemisphere
- B. Left cerebral hemisphere
- C. Cerebellum
- D. Brainstem

A. Motor tracts decussate, so deficits appear contralateral to the lesion.

Q15 · W2

Which cell predominates in acute inflammation?

- A. Neutrophils
- B. Macrophages
- C. Lymphocytes
- D. Eosinophils

A. Neutrophils arrive first. Macrophages and lymphocytes dominate chronic inflammation and drive fibrosis.

Q16 · W6

Which finding distinguishes DKA from HHS?

- A. Ketones with a low pH
- B. Elevated blood glucose
- C. Dehydration
- D. Altered mental status

A. Both have high glucose and dehydration. Only DKA has enough insulin deficiency to permit ketogenesis and acidosis.

Q17 · W1

Which adaptation describes an increase in the number of cells?

- A. Hyperplasia
- B. Hypertrophy
- C. Atrophy
- D. Metaplasia

A. Hyperplasia = more cells. Hypertrophy = bigger cells, and it is the only option for tissues that cannot divide.

Q18 · W5

Weakness that worsens with use and improves with rest suggests:

- A. Myasthenia gravis
- B. Muscular dystrophy
- C. Guillain-Barré syndrome
- D. Multiple sclerosis

A. Fatigability localises to the neuromuscular junction, where antibodies have reduced available ACh receptors.

Q19 · W3

Loss of both copies of p53 contributes to cancer because p53 normally:

- A. Arrests the cell cycle for repair and triggers apoptosis
- B. Stimulates cell division
- C. Promotes angiogenesis
- D. Transports oxygen

A. p53 is the guardian of the genome. Without it, damaged cells survive and accumulate further mutations.

Q20 · W1

A patient with prolonged vomiting is most likely to develop:

- A. Metabolic alkalosis
- B. Metabolic acidosis
- C. Respiratory acidosis
- D. Respiratory alkalosis

A. Vomiting loses gastric acid, raising pH. Diarrhoea loses bicarbonate and does the opposite.

Q21 · W4

Which describes the Cushing triad?

- A. Rising BP, falling pulse, irregular respirations
- B. Falling BP, rising pulse, rapid respirations
- C. Rising BP, rising pulse, apnoea
- D. Falling BP, falling pulse, tachypnoea

A. A late sign of raised ICP. Option B is hypovolaemic shock – the opposite pattern.

Q22 · W2

Anaphylaxis is mediated by:

- A. IgE bound to mast cells
- B. IgG against tissue antigens
- C. Immune complexes
- D. Cytotoxic T cells

A. Allergen cross-links IgE on mast cells, causing massive histamine release with airway compromise and distributive shock.

Q23 · W6

Hyperpigmentation in Addison disease results from:

- A. Elevated ACTH, which shares a precursor with melanocyte-stimulating hormone
- B. Excess cortisol
- C. High aldosterone
- D. Elevated glucose

A. The failing adrenal drives ACTH sky-high, and that same precursor pathway darkens the skin – especially creases and scars.

Q24 · W5

The priority assessment in Guillain-Barré syndrome is:

- A. Respiratory function
- B. Pupillary response
- C. Blood glucose
- D. Bowel sounds

A. Ascending paralysis reaching the diaphragm causes ventilatory failure, which is what kills in this condition.

Q25 · W1

Why does acidosis raise serum potassium?

- A. Hydrogen ions enter cells and potassium shifts out
- B. The kidneys retain potassium
- C. Aldosterone rises
- D. Potassium intake increases

A. It is a transcellular shift, not a true surplus "which is why correcting the acidosis can unmask profound depletion.

Q26 · W3

A chemotherapy patient develops ↑K, ↑phosphate, ↑uric acid, and ↓calcium. This is:

- A. Tumour lysis syndrome
- B. Hypercalcaemia of malignancy
- C. SIADH
- D. Addisonian crisis

A. Rapid tumour cell death spills intracellular contents; the released phosphate binds calcium, which is why calcium alone falls.

Q27 · W4

A lucid interval following head injury, then rapid deterioration, suggests:

- A. Epidural haematoma
- B. Concussion
- C. Subdural haematoma
- D. Diffuse axonal injury

A. Arterial bleeding fills the space quickly; the lucid interval occurs before intracranial compensation is exhausted.

Q28 · W6

Which is the greatest immediate danger in severe hypoglycaemia?

- A. The brain has no glucose reserve, so seizure and coma follow within minutes
- B. Ketoacid accumulation
- C. Osmotic diuresis
- D. Hyperkalaemia

A. Neurons depend on a continuous glucose supply. Hypoglycaemia is faster and more immediately lethal than hyperglycaemia.

Q29 · W2

Why can serious infection present with fever alone in a neutropenic patient?

- A. Without neutrophils there is no pus or local inflammation to localise the infection
- B. Fever suppresses other symptoms
- C. Bacteria grow more slowly
- D. Antibiotics mask the findings

A. The visible signs of infection are produced largely by neutrophils. With none present, infection can be advanced while appearing unimpressive.

Q30 · W1+W6

A patient in DKA breathes deeply and rapidly. This represents:

- A. Respiratory compensation for metabolic acidosis
- B. A primary respiratory alkalosis
- C. Respiratory acidosis
- D. Airway obstruction

A. Ketoacids consume bicarbonate and drop the pH; Kussmaul respirations eliminate CO_2 , to push it back up. A Week 1 principle explaining a Week 6 emergency.

 PLAN

Six-Day Study Plan

If you have a week, use it in this order — foundations first, because everything else is built on them.

Day 1 — Week 1

Foundations. Do 20 ABGs with the 4-step method until it is automatic. Learn the adaptation table and necrosis vs apoptosis cold. Topics 1–4.

Day 2 — Week 2

Immunity. Build the four-hypersensitivity table from memory, then the inflammation cascade. Topics 5–8.

Day 3 — Week 3

Oncology. Benign vs malignant, oncogene vs tumour suppressor, staging vs grading, and the oncologic emergencies. Topics 9–12.

Day 4 — Weeks 4 & 5

Neuro. The ICP cascade and stroke first, then meningitis CSF, the NMJ disorders, and neurogenic shock. Topics 13–20.

Day 5 — Week 6

Endocrine. Learn the feedback rule, then apply it to thyroid, adrenal, and ADH. Finish with DKA vs HHS. Topics 21–25.

Day 6 — Rehearsal

Read topics 26–27 (threads and confusion pairs) twice, then take the 30-question exam cold and timed. Re-study only what you miss.

THREE HABITS THAT RAISE SCORES ON THIS EXAM

1. **Answer the mechanism, not the disease.** When asked "which change explains," the right option is a *process* (ATP failure, receptor blockade, negative feedback), not a diagnosis.

2. **Find the discriminator before reading the options.** Decide what single fact separates the likely candidates, then look for the option containing it.
3. **Check whether the question asks "what is happening" or "what do I do first."** Priority questions almost always want **stop the harm** before assessment or documentation.

SCI 225 · Pathophysiology · Midterm Review Guide (Weeks 1–6)

Dr. Vidya Aggrawal · Nightingale College

Study aid for course review. Always follow current clinical guidelines and your instructor's stated exam scope.